I. Consider the following table **'StDetails'**:

StdId	FirstName	LastName	DOB
I01	Ankit	Verma	2001-04-22
I02	Sujeet	Kumar	2002-02-14
I03	Vikram	Singh	2001-07-02
I04	Bhisham	Sharma	2000-08-16

1. Write a query to create a table 'StDetails' with the fields shown in the above table. Use suitable data types for the fields and set StdId field as a primary key .
2. Write a query to insert the values shown above in the created table.
3. Write a query to Concatenate FirstName and LastName from the table 'StDetails'.
4. Write a query to show FirstName of the Students in uppercase from the table 'StDetails'.
5. Write a query to display all the details of students in descending order of the field LastName.
6. Write a query to display Current Date and Time.

II. Create a table **'Sports'** based on the following table instance chart:

Column Name	**CoachId**	**CoachName**	**Sports**	**Salary**
Data Type	varchar	Varchar	Varchar	Integer
Length	5	20	20	7
Key / Constraints	Primary Key	Not Null	--	--

III. Write a query to insert following records with relevant information, in the table.

CoachId	CoachName	Sports	Salary
Cr01	Kapil	Cricket	40,000
Bd101	Gopichand	Badminton	54,000
H101	Gurjeet Singh	Hockey	45,000
SMG01	Mayank	Swimming	4700,0
Bd102	Saina	Badminton	48,000
Bd103	Sindhu	Badminton	45,000

a. Write a query to find the sum of the salary paid to Coaches as 'Total Salary Paid' from the above table **'Sports'** .

b. Write a query to add a new feild **'DateOfApp'** with data type 'DATE' to the above created table.

c. Write a query to display all the details of coach whose name starts with alphabet 'G'.

d. Write a query to show total number of coaches belongs to sports 'Badminton' from the table **'Sports'.**

IV. Create a table **'Employee'** based on the following table instance chart:

Column Name	**EmpId**	**EmpName**	**Designation**	**Dept.**
Data Type	varchar	Varchar	Varchar	Varchar
Length	3	25	20	20
Key / Constraints	Primary Key	Not Null	--	--

a. Write a query to insert following records with relevant information, in the table.

EmpId	**EmpName**	**Designation**	**Dept**
E01	Rakesh	Manager	HR
E02	Subhash	Accountant	Accounts
E03	Subhiksha	PRO	PRO
E04	Vaani	Computer Operator	IT

b. Write a query to change the Dept. PRO to office Whose EmpId is 'E03' from the table **'Employee'.**

c. Write a query to delete a record of employee Subhash from the table **'Employee'.**

d. Write a query to show number of characters in the field Designation having EmpId as 'E04'.

e. Write a query to show current date.

V. Consider the following table **'Employee'**:

Empcode	**EmpName**	**DateofBirth**	**Salary**
I01	Jacob	1990-12-08	30,000
I02	Rahul	1992-03-18	50,000
I03	Sania	1990-04-22	45000
I04	Suyash	1991-02-19	22,000
I05	Tanya	1990-06-08	35,000
I06	Abhishek	1992-10-14	30,000

a. Write a query to create a table **'Employee'** with the fields shown in the above table. Use suitable data types for the fields and add a primary key to the table.

b. Write a query to display the details of employee whose salary is more than 30,000.

c. Write a query to display the name of employees whose name starts with **'S'**.
d. Write a query to display Empcode, Empname & salary from the table **'Employee'** in descending order of their names.
e. Write a query to join the empcode and empname field.
f. Write a query to display details of employee having minimum salary.
g. Write a query to find out square root of 82.

VI. Consider the following table **'Students'**:

AdmNo	Name	Stream	AvgMark
1001	Sunita	Arts	78
1002	Md. Ali	Science	89
1003	Manik Dev	Commerce	68
1004	Pavan Gupta	Science	73
1005	Sanjeev Verma	Commerce	90
1006	Shiv Kumar	Arts	75

a. Write a query to create a table **'Students'** with the fields shown in the above table. Use suitable data types for the fields.
b. Write a query to display the details of student whose Admission Number is 1003 and Stream is Commerce.
c. Write a query to display Names of students in capital letters.
d. Write a query to update the stream of the student whose AdmNo is 1004 to Commerce.
e. Write a query to add a new column **'Grade'** in the create table
f. Write a query to insert a new record in the table with relevant values.
g. Write a query to display the total of Average Marks from the above table.

Program Using Pandas and Matplotlib

I. Program Using Pandas

a. Write Python code to create a Series object **'Result'** storing percentage marks of four students with its index as 'Arun', 'Monika', 'Shyam' & 'Kiran'.
b. Write a code to create a DataFrame with following Data. (Assign any name for dataframe)

	Name	Marks
RollNo_1	Raj	70
RollNo_2	Simran	95
RollNo_3	Vicky	80

c. Write a program to print one row at a time from the above created dataframe.
d. Write a code to display total marks from the above created dataframe.
e. Write a program to export this dataframe to csv file. (You can choose any location)

II. Program using Matplotlib.

1. **'Perf'** is a list that stores marks of a student in 10 unit tests. Write a program to plot a line chart of the Students Performance in these 10 unit tests. Give appropriate axis labels. **[Maximum marks per unit test is 20]**

Program Using Pandas and Matplotlib

I. Program Using Pandas

a. Write Python code to create a Series object **'Emp'** storing names of 10 employees and write a code to display first 3 records and last 4 records from series object
b. Write a code to create a DataFrame with following Data. (Assign any name for dataframe)

	Population	**Schools**
Chennai	10927986	7925
Kolkata	12691836	6543
Mumbai	4631392	4589
Delhi	74135498	3765

c. Write a code to access the column Schools from the above created dataframe.
d. Write a code to add new column **'Hospital'** with relevant values.

II. Program using Matplotlib.

Write a program to plot a horizontal bar chart from the height of five students. Give appropriate chart title and axis labels.

Practice Paper - 1

SUBJECT – INFORMATICS PRACTICES (065)

M.M. – 30 **TIME ALLOWED- 3 Hrs.**

Program Using Pandas and Matplotlib (08 Marks)

III. Program Using Pandas **05**

1. Write Python code to create a Series object 'Temp2' storing temperatures of seven days of week. 1
   a) Name its index as 'Sunday', 'Monday', 'Tuesday'……'Saturday'. 1
2. Consider the following Data.

	Name	Marks
RollNo_1	Raj	70
RollNo_2	Simran	95
RollNo_3	Vicky	80

   a. Create a dataframe with the above data. 1
   b. Write a program to print Dataframe 'StMarks', one row at a time. 1
   c. Write a code to display minimum marks from the above created dataframe. 1

IV. Program using Matplotlib. **03**

2. Write a program to plot a line chart to depict the changing weekly price of potatoes for four weeks. Give appropriate axes labels.

SQL Queries (07 Marks)

1. Consider the following table **'Items'**:

Icode	Descp	Price	QOH	ROL
I01	Milk	15.00	20	10
I02	Cake	5.00	60	20
I03	Bread	9.00	40	10
I04	Biscuit	10.00	50	40
I05	Namkeen	15.00	100	50
I06	Cream Roll	7.00	10	20

7. Write a query to create a table 'Items' with the fields shown in the above table. Use suitable data types for the fields. 1
8. Write a query to display the field descp whose price is more than 10. 1
9. Write a query to display the descriptions that starts with 'B'. 1
10. Write a query to display Icode, Desc, & Price from the table in descending order of Descp. 1
11. Write a query to view the structure of the table. 1
12. Write a query to display details of items having minimum price. 1
13. Write a query to insert a new record in the table. 1

Practice Paper - 2

SUBJECT – INFORMATICS PRACTICES (065)

M.M. – 30 **TIME ALLOWED- 3 Hrs.**

Program Using Pandas and Matplotlib (08 Marks)

III. Program Using Pandas **05**

1. A series object (say T1) stores the average temperature recorded on each day of a month. Write code to display the temperatures recorded on:

(i) First 7 days 1

(ii) Last 7 days. **1**

2. Consider the following Dataframe 'mdf1'.

	Hospitals	Schools
Chennai	**189**	**7925**
Kolkata	**208**	**6543**
Mumbai	**149**	**4589**
Delhi	**157**	**3765**

1. Write a code to create the following dataframe. . 1
2. Write a code to access the column Schools from the above created dataframe. 1
3. Write a code to add new column 'Population' with relevant values. 1

I. Program using Matplotlib. **03**

1. Write a program to create bar chart on Student's performance in 10 Unit tests. Take the marks of students according to your choice. Give appropriate axes labels. [Maximum marks per unit test is 50]

SQL Queries (07 Marks)

3. Consider the following table **'Students'**:

AdmNo	Name	Stream	Optional Subject	Percentage
1001	Sunita	Arts	Hindi	65
1002	Md. Ali	Science	IP	69
1003	Manik Dev	Commerce	IP	71
1004	Pavan Gupta	Science	P.E.	62
1005	Sanjeev Verma	Commerce	IP	76
1006	Shiv Kumar	Arts	Hindi	68

h. Write a query to create a table 'Items' with the fields shown in the above table. Use suitable data types for the fields. 1
i. Which field can be taken as primary key and why? 1
j. Write a query to display the details of student whose AdmNo is 1003. 1
k. Write a query to display Icode, Desc, & Price from the table in ascending order of Name. 1
l. Write a query to update the optional subject of student whose AdmNo is 1006. 1
m. Write a query to insert a new record in the table. 1
n. Write a query to display maximum percentage of student from the above table. 1

Practice Paper - 3
SUBJECT – INFORMATICS PRACTICES (065)

M.M. – 30 **TIME ALLOWED- 3 Hrs.**

Program Using Pandas and Matplotlib (08 Marks)

I. Program Using Pandas **05**

1. Series objects Temp1, Temp2, Temp3, Temp4 store the temperatures of days of week1, week2, week3, weeken4 respectively. Write a script to create a series object with the mentioned names 1

(a) Print the average temperature per week 1

3. Consider the following Data.

	Name	Marks
RollNo_1	Raj	70
RollNo_2	Simran	95
RollNo_3	Vicky	80

a. Create a dataframe with the above data. 1
b. Write a program to change the marks of students to 75, 92, 83. 1
c. Write a code to display maximum marks from the above created dataframe. 1

1. Program using Matplotlib. **03**

Consider the following table:

Country	**Gold**	**Silver**	**Bronze**	**Total**
Australia	**80**	**59**	**59**	**198**
England	**45**	**45**	**60**	**136**
India	**26**	**20**	**20**	**66**
New Zealand	**15**	**40**	**27**	**82**

a. Write a program to plot a horizontal bar chart from the medals won by India. Give appropriate axes labels. 3

SQL Queries (07 Marks)

4. Consider the following table **'Items'**:

Icode	**Descp**	**Price**	**QOH**	**ROL**
I01	Milk	15.00	20	10
I02	Cake	5.00	60	20
I03	Bread	9.00	40	10

I04	Biscuit	10.00	50	40
I05	Namkeen	15.00	100	50
I06	Cream Roll	7.00	10	20

a. Write a query to create a database with name 'Adarsh'. 1
b. Write a query to create a table 'Items' with the fields shown in the above table. Use suitable data types for the fields. 1
c. Write a query to insert the values as shown in table in the table created. 1
d. Write a query to display the description in capital letters(Uppercase) 1
e. Write a query to add a new change the price value of Namkeen to 18.00 1
f. Write a query to remove the record of item having icode 'I06' from the table. 1
g. Write a query to display details of items having maximum price. 1

Practice Paper - 4
SUBJECT – INFORMATICS PRACTICES (065)

M.M. – 30 **TIME ALLOWED- 3 Hrs.**

Program Using Pandas and Matplotlib (08 Marks)

I. Program Using Pandas **05**

1. Given a Series 'Ser1' that stores the area of of some states in km^2.

TotSales=pd.Series([34567, 890, 450, 67892, 34677, 78902, 256711, 678291, 637632, 25723, 2367, 11789, 345, 256517])

a) Write code to find out the smallest four areas from the given series. 1

b) Write code to find out the biggest four areas from the given series. 1

.2. Write a program to create a Dataframe storing salesmen details (name, zone, sales) of five salesmen. 2.

1. Program using Matplotlib. **03**

2. Consider the following table:

Country	Gold	Silver	Bronze	Total
Australia	80	59	59	198
England	45	45	60	136
India	26	20	20	66
New Zealand	15	40	27	82

a. Write a program to plot a line chart of Australia and India from the above given data. 2

b. The line style for India should be dashed type. 1

SQL Queries (07 Marks)

1. Write a query to Create a table 'Employee' with the data given. 1

Column Name	ID	First_Name	Last_Name	Dept_Id
Key type	Primary Key			
Data type	Integer	Varchar	Varchar	Varchar
Length	5	25	20	4

2. Write a query to Insert 4 records to the table created. 1
3. Write a query to add a new column Salary of datatype Integer and length (7). 1
4. Write a query to enter the values in the column created i.e. Salary.
5. Write a query to drop the column Dept_id. 1
6. Write a query to display Id, First_Name, and Salary from the table 'Employee' whose name contains 'A' as third alphabet. 1
7. Write a query to display the average salary. 1

Program Using Pandas and Matplotlib (08 Marks)

V. Program Using Pandas **05**

4. Write Python code to create a Series object 'Temp2' storing temperatures of seven days of week. 1
   b) Name its index as 'Sunday', 'Monday', 'Tuesday'……'Saturday'. 1
5. Consider the following Data.

	Name	Marks
RollNo_1	Raj	70
RollNo_2	Simran	95
RollNo_3	Vicky	80

   d. Create a dataframe with the above data. 1
   e. Write a program to print Dataframe 'StMarks', one row at a time. 1
   f. Write a code to display minimum marks from the above created dataframe. 1

VI. Program using Matplotlib. **03**

3. Write a program to plot a line chart to depict the changing weekly price of potatoes for four weeks. Give appropriate axes labels.

SQL Queries (07 Marks)

5. Consider the following table **'Items'**:

Icode	Descp	Price	QOH	ROL
I01	Milk	15.00	20	10
I02	Cake	5.00	60	20
I03	Bread	9.00	40	10
I04	Biscuit	10.00	50	40
I05	Namkeen	15.00	100	50
I06	Cream Roll	7.00	10	20

14. Write a query to create a table 'Items' with the fields shown in the above table. Use suitable data types for the fields. 1
15. Write a query to display the field descp whose price is more than 10. 1
16. Write a query to display the descriptions that starts with 'B'. 1
17. Write a query to display Icode, Desc, & Price from the table in descending order of Descp. 1
18. Write a query to view the structure of the table. 1
19. Write a query to display details of items having minimum price. 1
20. Write a query to insert a new record in the table. 1

Program Using Pandas and Matplotlib (08 Marks)

IV. Program Using Pandas **05**

1. A series object (say T1) stores the average temperature recorded on each day of a month. Write code to display the temperatures recorded on:
(i) First 7 days 1
(ii) Last 7 days. **1**

6. Consider the following Dataframe 'mdf1'.

	Hospitals	**Schools**
Chennai	**189**	**7925**
Kolkata	**208**	**6543**
Mumbai	**149**	**4589**
Delhi	**157**	**3765**

4. Write a code to create the following dataframe. . 1
5. Write a code to access the column Schools from the above created dataframe. 1
6. Write a code to add new column 'Population' with relevant values. 1

II. Program using Matplotlib. **03**

2. Write a program to create bar chart on Student's performance in 10 Unit tests. Take the marks of students according to your choice. Give appropriate axes labels. [Maximum marks per unit test is 50]

SQL Queries (07 Marks)

7. Consider the following table **'Students'**:

AdmNo	Name	Stream	Optional Subject	Percentage
1001	Sunita	Arts	Hindi	65
1002	Md. Ali	Science	IP	69
1003	Manik Dev	Commerce	IP	71
1004	Pavan Gupta	Science	P.E.	62
1005	Sanjeev Verma	Commerce	IP	76
1006	Shiv Kumar	Arts	Hindi	68

21. Write a query to create a table 'Items' with the fields shown in the above table. Use suitable data types for the fields. 1
22. Which field can be taken as primary key and why? 1
23. Write a query to display the details of student whose AdmNo is 1003. 1
24. Write a query to display Icode, Desc, & Price from the table in ascending order of Name. 1
25. Write a query to update the optional subject of student whose AdmNo is 1006. 1
26. Write a query to insert a new record in the table. 1
27. Write a query to display maximum percentage of student from the above table. 1

Program Using Pandas and Matplotlib (08 Marks)

II. Program Using Pandas **05**

1. Series objects Temp1, Temp2, Temp3, Temp4 store the temperatures of days of week1, week2, week3, weeken4 respectively. Write a script to create a series object with the mentioned names 1

(a) Print the average temperature per week 1

6. Consider the following Data.

	Name	Marks
RollNo_1	Raj	70
RollNo_2	Simran	95
RollNo_3	Vicky	80

g. Create a dataframe with the above data. 1
h. Write a program to change the marks of students to 75, 92, 83. 1
i. Write a code to display maximum marks from the above created dataframe. 1

2. Program using Matplotlib. **03**

Consider the following table:

Country	Gold	Silver	Bronze	Total
Australia	**80**	**59**	**59**	**198**
England	**45**	**45**	**60**	**136**
India	**26**	**20**	**20**	**66**
New Zealand	**15**	**40**	**27**	**82**

b. Write a program to plot a horizontal bar chart from the medals won by India. Give appropriate axes labels. 3

SQL Queries (07 Marks)

8. Consider the following table **'Items'**:

Icode	Descp	Price	QOH	ROL
I01	Milk	15.00	20	10
I02	Cake	5.00	60	20

I03	Bread	9.00	40	10
I04	Biscuit	10.00	50	40
I05	Namkeen	15.00	100	50
I06	Cream Roll	7.00	10	20

28. Write a query to create a database with name 'Adarsh'. 1
29. Write a query to create a table 'Items' with the fields shown in the above table. Use suitable data types for the fields. 1
30. Write a query to insert the values as shown in table in the table created. 1
31. Write a query to display the description in capital letters(Uppercase) 1
32. Write a query to add a new change the price value of Namkeen to 18.00 1
33. Write a query to remove the record of item having icode 'I06' from the table. 1
34. Write a query to display details of items having maximum price. 1

Program Using Pandas and Matplotlib (08 Marks)

II. Program Using Pandas **05**

2. Given a Series 'Ser1' that stores the area of of some states in km^2.

TotSales=pd.Series([34567, 890, 450, 67892, 34677, 78902, 256711, 678291, 637632, 25723, 2367, 11789, 345, 256517])

c) Write code to find out the smallest four areas from the given series. 1

d) Write code to find out the biggest four areas from the given series. 1

.2. Write a program to create a Dataframe storing salesmen details (name, zone, sales) of five salesmen. 2.

3. Program using Matplotlib. **03**

4. Consider the following table:

Country	Gold	Silver	Bronze	Total
Australia	80	59	59	198
England	45	45	60	136
India	26	20	20	66
New Zealand	15	40	27	82

c. Write a program to plot a line chart of Australia and India from the above given data. 2

d. The line style for India should be dashed type. 1

SQL Queries (07 Marks)

8. Write a query to Create a table 'Employee' with the data given. 1

Column Name	ID	First_Name	Last_Name	Dept_Id

Key type	Primary Key			
Data type	Integer	Varchar	Varchar	Varchar
Length	5	25	20	4

9. Write a query to Insert 4 records to the table created. 1
10. Write a query to add a new column Salary of datatype Integer and length (7). 1
11. Write a query to enter the values in the column created i.e. Salary.
12. Write a query to drop the column Dept_id. 1
13. Write a query to display Id, First_Name, and Salary from the table 'Employee' whose name contains 'A' as third alphabet. 1
14. Write a query to display the average salary. 1